Lecture 8

L'Hopital Rule (Indeterminate Form)

- **$\frac{0}{0}, \frac{\infty}{\infty}$ form:**

If $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f(a)}{g(a)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$

Then, $\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$ [Do it until the indeterminate form goes away.]

- **$\frac{0}{0}$ form:**

(1) $\lim_{x \to 0} \frac{\sin 2x}{x}$

$= \lim_{x \to 0} \frac{2\cos 2x}{1}$

$= \lim_{x \to 0} 2\cos 2x$

$= 2 \cdot 1$

$= 2$

(2) $\lim_{x \to \pi} \frac{\sin x}{x-\pi}$

$= \lim_{x \to \pi} \frac{\cos x}{1}$

$= \cos \pi$

$= -1$

(3) $\lim_{x \to 3} \frac{x-3}{3x^2-13x+12}$

$= \lim_{x \to 3} \frac{1}{6x-13}$

$= \frac{1}{6 \cdot 3-13}$

$= \frac{1}{18-13}$

$= \frac{1}{5}$

(4) $\lim_{x \to 1} \frac{\ln x}{x-1}$

$= \lim_{x \to 1} \frac{\frac{1}{x}}{1-0}$

$= \lim_{x \to 1} \frac{1}{x}$

$= \frac{1}{1}$

$= 1$

We know,

1 $\lim_{x \to 0} \frac{\sin x}{x} = 1$

2. $\lim_{x \to 0} \frac{\sin(ax)}{ax} = 1$

(5) $\lim_{x \to 0} \frac{\sin 2x}{\sin 5x}$

$= \lim_{x \to 0} \frac{2\cos 2x}{5\cos 5x}$

$= \frac{2 \cdot 1}{5 \cdot 1}$

$= \frac{2}{5}$

Alternative: $\lim_{x \to 0} \frac{\sin 2x}{\sin 5x}$

$$= \frac{\lim_{x \to 0} \frac{\sin 2x}{2x} \times 2x}{\lim_{x \to 0} \frac{\sin 5x}{5x} \times 5x}$$

$$= \frac{\lim_{x \to 0} \frac{\sin 2x}{2x}}{\lim_{x \to 0} \frac{\sin 5x}{5x}} \times \frac{2x}{5x}$$

$= \frac{1}{1} \times \frac{2}{5}$ $\left[\lim_{x \to 0} \frac{\sin(ax)}{ax} = 1\right]$

$= \frac{2}{5}$

- $\frac{\infty}{\infty}$ **form:**

(1) $\lim\limits_{x \to +\infty} \frac{e^{3x}}{x^2}$

$= \lim\limits_{x \to +\infty} \frac{3e^{3x}}{2x}$

$= \lim\limits_{x \to +\infty} \frac{9e^{3x}}{2}$ $\left[\frac{\infty}{2} \text{ so we will not differentiate anymore}\right]$

$= \frac{9}{2} \lim\limits_{x \to +\infty} e^{3x}$

$= +\infty$

(2) $\lim\limits_{x \to +\infty} \frac{x}{e^x}$

$= \lim\limits_{x \to +\infty} \frac{1}{e^x}$

$= 0$ $\left[\text{As } x \to \infty, e^x \to \infty, \text{so}, \frac{1}{e^x} \to 0\right]$

- $\mathbf{0 \times \infty}$ **form:**

(1) $\lim\limits_{x \to \pi} (x - \pi)\cot x$

$= \lim\limits_{x \to \pi} \frac{x-\pi}{\frac{1}{\cot x}}$

$= \lim\limits_{x \to \pi} \frac{x-\pi}{\tan x}$

$= \lim\limits_{x \to \pi} \frac{1}{\sec^2 x}$

$= \lim\limits_{x \to \pi} \cos^2 x$

$= \left(\lim\limits_{x \to \pi} \cos x\right)^2$

$= (\cos \pi)^2$

$= (-1)^2$

$= 1$

- $\infty \pm \infty$ **form:**

(1) $\lim\limits_{x \to 0} \left(\frac{1}{x} - \frac{1}{\sin x}\right)$

$= \lim\limits_{x \to 0} \frac{\sin x - x}{x \sin x}$

$= \lim\limits_{x \to 0} \frac{\cos x - 1}{\sin x + x\cos x}$ $\quad [\frac{0}{0}$ form$]$

$= \lim\limits_{x \to 0} \frac{-\sin x}{\cos x + \cos x - x\sin x}$

$= \frac{-0}{1+1-0}$

$= 0$

H.W. Exercise 3.6 (7-26, 33-38)

- $0^0, \infty^0, 1^\infty$ **form:**

(1) $\lim\limits_{x \to 0} (1+x)^{\frac{1}{x}} = ?$

Let, $y = (1+x)^{\frac{1}{x}}$

$\Rightarrow \ln y = \ln(1+x)^{\frac{1}{x}}$

$\Rightarrow \ln y = \frac{1}{x}\ln(1+x)$

$\Rightarrow \ln y = \frac{\ln(1+x)}{x}$

$\Rightarrow \lim\limits_{x \to 0} \ln y = \lim\limits_{x \to 0} \frac{\ln(1+x)}{x}$ $\quad [\frac{0}{0}$ form$]$ $\qquad \left[\frac{\ln(1+0)}{0} = \frac{\ln 1}{0} = \frac{0}{0}\right]$

$\Rightarrow \lim\limits_{x \to 0} \ln y = \lim\limits_{x \to 0} \frac{\frac{1}{1+x} \times 1}{1}$

$\Rightarrow \lim\limits_{x \to 0} \ln y = \lim\limits_{x \to 0} \frac{1}{1+x}$

$\Rightarrow \lim\limits_{x \to 0} \ln y = \frac{1}{1+0} = 1$

$\Rightarrow \lim\limits_{x \to 0} \ln y = 1$

$\therefore \lim\limits_{x \to 0} (\ln y) = 1$

$\Rightarrow \lim_{x \to 0} y = e^1$ [If $\ln a = b$, then $a = e^b$]

$\therefore \lim_{x \to 0} (1+x)^{\frac{1}{x}} = e$

(2) $\lim_{x \to 0} (e^x + x)^{\frac{1}{x}} = ?$

Let, $y = (e^x + x)^{\frac{1}{x}}$

$\Rightarrow \ln y = \ln(e^x + x)^{\frac{1}{x}}$

$\Rightarrow \ln y = \frac{1}{x} \ln(e^x + x)$

$\Rightarrow \ln y = \frac{\ln(e^x + x)}{x}$

Now, $\lim_{x \to 0} \ln y = \lim_{x \to 0} \frac{\ln(e^x+x)}{x}$ [$\frac{0}{0}$ form]

$\Rightarrow \lim_{x \to 0} \ln y = \lim_{x \to 0} \frac{\frac{1}{e^x + x} \times (e^x + 1)}{1}$

$\Rightarrow \lim_{x \to 0} \ln y = \lim_{x \to 0} \frac{e^x + 1}{e^x + x}$

$\Rightarrow \lim_{x \to 0} \ln y = \frac{1+1}{1+0}$

$\Rightarrow \lim_{x \to 0} \ln y = \frac{2}{1}$

$\Rightarrow \lim_{x \to 0} \ln y = 2$

$\therefore \lim_{x \to 0} (\ln y) = 2$

$\Rightarrow \lim_{x \to 0} y = e^2$ [If $\ln a = b$, then $a = e^b$]

$\therefore \lim_{x \to 0} (e^x + x)^{\frac{1}{x}} = e^2$

H.W. Exercise 3.6 (27-32, 39-45)

Example: $\lim\limits_{x \to +\infty} \frac{(x+1)^x}{x^x} = \lim\limits_{x \to +\infty} \left(\frac{x+1}{x}\right)^x = \lim\limits_{x \to +\infty} \left(1+\frac{1}{x}\right)^x = ?$

Solution: Let, $p = \frac{1}{x}$, so $x = \frac{1}{p}$; as $x \to +\infty$, $p \to 0$.

Now, we get,

$$\lim_{p \to 0} (1+p)^{\frac{1}{p}} = ?$$

Let, $y = (1+p)^{\frac{1}{p}}$

$$\Rightarrow \ln y = \frac{1}{p}\ln(1+p)$$

$$\Rightarrow \lim_{p \to 0} \ln y = \lim_{p \to 0} \frac{\ln(1+p)}{p} \quad [\frac{0}{0} \text{ form}]$$

$$\Rightarrow \lim_{p \to 0} \ln y = \lim_{p \to 0} \frac{\frac{1}{1+p}}{1}$$

$$\Rightarrow \lim_{p \to 0} \ln y = \lim_{p \to 0} \frac{1}{1+p}$$

$$\frac{d}{dp}(\ln(1+p)) = \frac{1}{1+p}$$

$$\frac{d}{dp}(p) = 1$$

$$\Rightarrow \lim_{p \to 0} \ln y = \frac{1}{1+0}$$

$$\Rightarrow \lim_{p \to 0} \ln y = 1$$

$$\therefore \lim_{p \to 0} \ln y = 1$$

$$\Rightarrow \lim_{p \to 0} y = e^1 \qquad [\text{If } \ln a = b, \text{ then } a = e^b]$$

$$\Rightarrow \lim_{p \to 0} (1+p)^{\frac{1}{p}} = e$$

$$\therefore \lim_{x \to +\infty} \left(1+\frac{1}{x}\right)^x = e$$

H.W. Exercise 1.3 (39-40)

H.W. Practice Sheet All Maths